INCH-POUND

MIL-T-46632B(OS)
11 August 1994
SUPERSEDING
MIL-T-46632A(MU)
20 August 1969

MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M2A2 METAL PARTS ASSEMBLY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Departments of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M2A2 Cartridge Actuated Thruster Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures And Tables For Inspection By Attributes

MIL-STD-109 Quality Assurance Terms And Definitions

MIL-STD-413 Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

8593874	Thruster, Cartridge Actuated, M2A2 Metal Parts Assembly
8797633	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M2A2
8797634	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M2A2
9235819	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M2A2

U.S. Air Force

TPO 00-921-5048	Thruster, Cartridge Actuated, M2A2

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.4 and 6.3).

3.2 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

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3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure. The body and head shall withstand a hydrostatic pressure of 10,000 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.4.1).

3.5 Torque.

3.5.1 Assembly torque. The body and holding rings shall be assembled hand tight. A torque of 100 + 20 inch pounds shall then be applied between the body and holding ring for the purpose of tightening the assembly prior to spotting threads for setscrew installation.

3.5.2 Breakaway torque. The body with respect to the holding ring shall withstand a breakaway torque application of 400 inch pounds, minimum, without any movement or deformation.

3.6 Structural strength. The assembled thruster shall withstand a static tensile force of 2,000 pounds, minimum, for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure (see 6.4.1).

3.7 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 55 ± 4 pounds is applied at the steady rate of 0.125 inch per minute.

3.8 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing and the requirements of 3.8.1 through 3.8.3.

3.8.1 Head and body. The head and body shall be assembled hand tight.

3.8.2 Piston locking spring and end sleeve. The piston locking spring shall enter the spring hole in the piston without any bending along its entire length. The assembly of the piston locking spring to the end sleeve, with the cap screw and washer, shall be hand tight. If axial misalignment of the piston locking spring occurs upon completion of assembly, it will be permissible to back off the cap screw sufficiently to eliminate axial misalignment of the piston locking spring.

3.8.3 End sleeve rotation. Upon completion of assembly and after the structural strength test, the end sleeve shall be free to rotate 360 degrees in a clockwise direction.

3.9 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

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3.10 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.7.

3.11 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.11.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.11.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77°F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Key, Drawing 8593857, covering a detail of Drawing 8593874.

Characteristics	Defect	Method of Inspection

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Critical
None defined

Major

101	Radial profile of length, inside and outside	Gage
102	Radial profile of width	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Surface finish improper	Visual
106	Burr or sharp corner	Visual
107	Foreign matter	Visual

Minor

201	Total length	Gage
202	Radii incorrect, except as otherwise classified	Gage

4.5.2.2 Trunnion, Drawing 8593860, covering a detail of Drawing 8593874.

Critical
None defined

Major

101	Concentricity of lugs	Gage
102	Symmetry of lugs with inside	Gage
103	Symmetry of flats at root of lugs with inside	Gage
104	Symmetry of lugs with sides	Gage
105	Perpendicularity of sides with inside	Gage
106	Diameter of lugs, min	Gage
107	Width	Gage
108	Width across flats, min	Gage
109	Thickness of wall, min	Gage
110	Radius at base of lugs incorrect	Gage
111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual

Minor

201	Diameter of lugs, max	Gage
202	Diameter, inside	Gage
203	Length of lugs	Gage
204	Width across flats, max	Gage
205	Thickness of wall, max	Gage
206	Location of lugs	Gage
207	Marking missing, incorrect, or illegible	Visual
208	Radius or chamfer incorrect, except as otherwise classified	Gage
209	Surface finish improper	Visual

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210	Burr or sharp corner	Visual
211	Foreign matter	Visual

4.5.2.3 Plug, Piston, Drawing 8593868, covering a detail of Drawing 8593874.

Critical
None defined

Major

101	Diameter	Gage
102	Metal defective	Visual

Minor

201	Thickness	Gage
202	Chamfer incorrect	Gage
203	Protective finish inadequate or defective	Visual
204	Surface finish improper	Visual
205	Burr or sharp corner	Visual
206	Foreign matter	Visual

4.5.2.4 Sleeve, End, Drawing 8593869, covering a detail of Drawing 8593874.

Critical
None defined

Major

101	Pitch diameter of external thread, min	Gage
102	Major diameter of external thread, min	Gage
103	Concentricity of cavity with body	Gage
104	Concentricity of shoulder with body	Gage
105	Symmetry of elongated slots with body	Gage
106	Diameter of body, min	Gage
107	Diameter of cavity, max	Gage
108	Diameter of shoulder, min	Gage
109	Width of slots	Gage
110	Location of slots	Gage
111	Radius of thread end	Gage
112	Thread damaged or not full	Visual
113	Metal defective	Visual
114	Protective finish inadequate or defective	Visual
115	Burr or sharp corner	Visual
116	Foreign matter	Visual

Minor

201	Pitch diameter of external thread, max	Gage
202	Pitch diameter of internal thread	Gage
203	Minor diameter of internal thread, max	Gage

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204	Diameter of body, max	Gage	
205	Diameter of cavity, min	Gage	
206	Diameter of shoulder, max	Gage	
207	Diameter of flange	Gage	
208	Diameter of thread undercut, min		Gage
209	Length of slot	Gage	
210	Variation in alignment, flange face with face of cavity seat		Gage
211	Depth of effective internal threads, min		Gage
212	Depth of small threaded hole, max		Gage
213	Total length	Gage	
214	Width across flats	Gage	
215	Width of shoulder	Gage	
216	Width of thread undercut	Gage	
217	Thickness of flange	Gage	
218	Location of flange	Gage	
219	Thickness of flange at flats		Gage
220	Location of flats	Gage	
221	Marking missing, incorrect, or illegible		Visual
222	Radius or chamfer, or both, incorrect, except as otherwise classified		Gage
223	Surface finish improper	Visual	

4.5.2.5 Spring, Piston, Locking, Drawing 8593871, covering a detail of Drawing 8593874.

Critical
None defined

Major

101	Load at specified heights	Gage	
102	Height solid, max	Gage	
103	Ends incorrect	Visual	
104	Protective finish inadequate or defective		Visual
105	Burr	Visual	
106	Foreign matter	Visual	

Minor

201	Diameter outside, closed, max		Gage
202	Inside diameter of eye	Gage	

4.5.2.6 Pin, Shear, Drawing 8593876, covering a detail of Drawing 8593874.

Critical
None defined

Major

101	Diameter, max	Gage	
102	Total length, min	Gage	

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103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual

Minor

201	Total length, max	Gage
202	Profile of ends	Visual
203	Surface finish improper	Visual

4.5.2.7 Piston, Drawing 8593877, covering a detail of Drawing 8593874.

Critical
None defined

Major

101	Concentricity of O-ring groove with body, large end	Gage
102	Concentricity of body, small end with body, large end	Gage
103	Concentricity of counterbore with body, large end	Gage
104	Concentricity of midsection with body, large end	Gage
105	Concentricity of locking groove with body, small end	Gage
106	Concentricity of small cavity with body, small end	Gage
107	Concentricity of key escape groove with body, small end	Gage
108	Concentricity of ring lock groove with midsection	Gage
109	Concentricity of body pilot with body, large end	Gage
110	Diameter of body, large end, min	Gage
111	Diameter of body pilot, large end, min	Gage
112	Diameter of O-ring groove	Gage
113	Diameter of shoulder, large end	Gage
114	Diameter of midsection	Gage
115	Diameter of body, small end	Gage
116	Diameter of key escape groove, max	Gage
117	Diameter of key locking groove, min	Gage
118	Diameter of small cavity	Gage
119	Diameter of ring lock groove	Gage
120	Diameter of counterbore	Gage
121	Length to body, large end, min	Gage
122	Length of pilot, small end	Gage
123	Location of angle in key groove	Gage
124	Depth to drill point, max	Gage
125	Length to end of key escape groove	Gage
126	Length of body, small end, min	Gage
127	Profile of key escape and locking groove	Gage
128	Profile of O-ring groove	Gage
129	Metal defective	Visual

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130	Protective finish inadequate or defective		Visual
131	Surface finish on body, large end O-ring groove improper		Visual
132	Burr or sharp corner	Visual	
133	Foreign matter	Visual	

Minor

201	Diameter of body pilot, large end, max		Gage
202	Diameter of body, large end, max		Gage
203	Diameter of key escape groove, min		Gage
204	Diameter of key locking groove, max		Gage
205	Length to body, large end, max		Gage
206	Diameter of large cavity	Gage	
207	Depth of counterbore	Gage	
208	Depth of small cavity	Gage	
209	Depth of long cavity	Gage	
210	Length to midsection, max	Gage	
211	Total length	Gage	
212	Width of shoulder, large end		Gage
213	Width of ring lock groove	Gage	
214	Width of O-ring groove	Gage	
215	Location of O-ring groove	Gage	
216	Location of ring lock groove		Gage
217	Radius or chamfer incorrect		Gage
218	Surface finish improper, except as otherwise classified		Visual

4.5.2.8 Pin, Firing, Drawing 8593878, covering a detail of Drawing 8593874.

Critical
None defined

Major

101	Concentricity of groove with body,		Gage
102	Concentricity of striker with body		Gage
103	Perpendicularity of striker face with body		Gage
104	True position of pin hole	Gage	
105	Diameter of body, min	Gage	
106	Diameter of groove,	Gage	
107	Diameter of striker	Gage	
108	Diameter of pin hole, max	Gage	
109	Depth of pin hole, max	Gage	
110	Length of striker, min	Gage	
111	Profile of groove	Gage	
112	Metal defective	Gage	
113	Protective finish inadequate or defective		Visual
114	Surface finish on body improper		Visual
115	Burr or sharp corner	Visual	
116	Foreign matter	Visual	

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Minor			
201	Diameter of body, max	Gage	
202	Diameter of pin hole, min	Gage	
203	Depth of pin hole, min	Gage	
204	Total length	Gage	
205	Length of striker, max	Gage	
206	Location of pin hole	Gage	
207	Width of groove	Gage	
208	Location of groove	Gage	
209	Depth of slot in base	Gage	
210	Width of slot in base	Gage	
211	Location of slot in base	Visual	
212	Marking missing, incorrect, or illegible		Visual
213	Radius incorrect	Gage	
214	Surface finish improper, except as otherwise classified		Visual

4.5.2.9 Ring, Snap, Drawing 8593879, covering a detail of Drawing 8593874.

Critical
None defined

Major			
101	Diameter inside	Gage	
102	Width	Gage	
103	Thickness of wall	Gage	
104	Metal defective	Visual	
105	Protective finish inadequate or defective		Visual
106	Surface finish improper	Visual	
107	Burr or sharp corner	Visual	
Minor			
201	Opening between ends	Gage	
202	Radius incorrect	Gage	
203	Foreign matter	Visual	

4.5.2.10 Ring, Holding, Drawing 8593880, covering a detail of Drawing 8593874.

Critical
None defined

Major			
101	Metal defective	Visual	
102	Protective finish inadequate or defective		Visual

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Minor

201	Pitch diameter of large thread		Gage
202	Minor diameter of large thread, max		Gage
203	Pitch diameter of small thread		Gage
204	Minor diameter of small thread		Gage
205	Location of holes	Gage	
206	Diameter	Gage	
207	Width across flats	Gage	
208	Width	Gage	
209	Thickness of wall at flat, min		Gage
210	Radius or chamfer incorrect		Gage
211	Surface finish improper	Visual	
212	Marking missing, incorrect, or illegible		Visual
213	Burr	Visual	
214	Foreign matter	Visual	

4.5.2.11 Head, Drawing 8593882, covering a detail of Drawing 8593874.

Critical
None defined

Major

101	Pitch diameter of thread, inlet port, max		Gage
102	Minor diameter of thread, inlet port, max		Gage
103	Pitch diameter of large thread, max		Gage
104	Minor diameter of large thread, max		Gage
105	Concentricity of midsection cavity with large thread		Gage
106	Perpendicularity of boss face with cavity		Gage
107	True position of pin hole	Gage	
108	Diameter of cavity at midsection, max		Gage
109	Depth of counterbore, small end		Gage
110	Length of effective thread, small end, min		Gage
111	Length of effective thread, large end, min		Gage
112	Length of thread, small end, max		Gage
113	Height of internal boss	Gage	
114	Angle of chamfer at small thread		Gage
115	Thread damaged or not full	Visual	
116	Metal defective	Visual	
117	Protective finish inadequate or defective		Visual
118	Surface finish of midsection cavity, small cavity counterbore, or chamfer, improper		Visual
119	Burr	Visual	
120	Foreign matter	Visual	

Minor

201	Pitch diameter of small thread, min		Gage
202	Pitch diameter of large thread, min		Gage

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203	Pitch diameter of setscrew thread	Gage
204	Minor diameter of setscrew thread, max	Gage
205	Diameter of counterbore, small end	Gage
206	Diameter of hole between midsection and small end	Gage
207	Diameter of cavity at midsection, min	Gage
208	Diameter of pin hole	Gage
209	Depth of large cavity	Gage
210	Depth of cavity, small end	Gage
211	Depth of setscrew cavity, max	Gage
212	Depth of midsection cavity from boss	Gage
213	Length of effective setscrew thread, min	Gage
214	Total length	Gage
215	Location of pin hole	Gage
216	Diameter of chamfer, small end	Gage
217	Diameter of large end	Gage
218	Diameter of small end	Gage
219	Diameter at midsection	Gage
220	Diameter of boss	Gage
221	Length to large body diameter	Gage
222	Length of small body diameter	Gage
223	Length of flat	Gage
224	Width across flats	Gage
225	Location of flats	Gage
226	Angle of internal boss incorrect	Visual
227	Marking missing, incorrect, or illegible	Visual
228	Radius or chamfer incorrect, except as otherwise classified	Gage
229	Surface finish improper, except as otherwise classified	Visual

4.5.2.12 Body, Drawing 8593883, covering a detail of Drawing 8593874.

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of small bore with long bore	Gage
104	Concentricity of key groove with small bore	Gage
105	Concentricity of long bore with cavity	Gage
106	Concentricity of counterbore with cavity	Gage
107	Concentricity of pitch diameter of thread with cavity	Gage
108	Concentricity of ring groove with small bore	Gage
109	Concentricity of counterbore with small bore	Gage
110	Diameter of cavity, max	Gage
111	Diameter of counterbore, thread end	Gage
112	Diameter of long bore, max	Gage
113	Diameter of key groove	Gage

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114	Diameter of small bore	Gage	
115	Diameter of counterbore, small end		Gage
116	Diameter of shoulder	Gage	
117	Location of key groove	Gage	
118	Straightness of long bore	Gage	
119	Profile of key groove	Gage	
120	Thread damaged or not full	Visual	
121	Metal defective	Visual	
122	Protective finish inadequate or defective		Visual
123	Surface finish of long bore improper		Visual
124	Burr	Visual	
125	Sharp corner, as specified	Visual	
126	Foreign matter	Visual	
	Minor		
201	Pitch diameter of thread, max		Gage
202	Diameter of cavity, min	Gage	
203	Diameter of long bore, min	Gage	
204	Diameter of thread pilot	Gage	
205	Diameter of ring groove	Gage	
206	Diameter of body, max	Gage	
207	Depth of counterbore, thread end		Gage
208	Depth of counterbore, small end		Gage
209	Total length	Gage	
210	Length of thread pilot	Gage	
211	Length to base of long bore	Gage	
212	Length to flange	Gage	
213	Width of ring groove	Gage	
214	Thickness of wall, long bore		Gage
215	Thickness of wall, internal undercut, if applicable		Gage
216	Location of thread undercut		Gage
217	Location of ring groove	Gage	
218	Diameter of thread undercut, min		Gage
219	Diameter of flange	Gage	
220	Depth of cavity	Gage	
221	Length of flats	Gage	
222	Length to start of flange angle, min		Gage
223	Length to end of flange angle, max		Gage
224	Width of undercut thread	Gage	
225	Width across flats	Gage	
226	Width of internal undercut, max, if applicable		Gage
227	Marking missing, incorrect, or illegible		Visual
228	Radius or chamfer incorrect		Gage
229	Surface finish improper, except as otherwise classified		Visual

4.5.2.13 Piston and Plug Assembly, Drawing 8593881, covering a detail of Drawing 8593874.

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Critical
None defined

Major
101 Diameter at plug end, max Gage

Minor
201 Marking missing, incorrect, or illegible Visual
202 Foreign matter Visual

4.5.2.14 Body and Pellet Assembly, Drawing 8593884, covering a detail of Drawing 8593874.

Critical
None defined

Major
101 Protrusion of pellets over thread, max Gage
102 Pellets loose, missing, or improperly inserted Visual
103 Thread damaged Visual
104 Burr Visual
105 Foreign matter Visual

Minor
201 Location of pellets Gage

4.5.2.15 End Sleeve Subassembly, consisting of the following components: Drawings 8593869, 8593871, 8593885, and MS35337-79, covering a detail of Drawing 8593874.

Critical
None defined

Major
101 Misalignment or bending of spring Visual

Minor
201 Any part missing Visual
202 Thread damaged or not full Visual
203 Burr Visual
204 Foreign matter Visual

4.5.2.16 Metal Parts Assembly, Drawing 8593874.

Critical
None defined

Major
101 End sleeve not free to rotate 360 degrees clockwise Manual
102 Any part missing, except as otherwise classified Visual

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103	Improper assembly	Visual
104	Thread damaged or not full	Visual
105	Marking missing, incorrect, or illegible	Visual
106	Burr	Visual
107	Foreign matter	Visual

Minor

201	Protective cap or plug missing	Visual
202	Protective finish inadequate or defective	Visual

4.5.2.17 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8797633.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.5.2.18 Box, Packing, Ammunition, Drawing 8797634.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.19 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9235819.

Critical
None defined

Major

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None defined

Minor		
201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each body and head shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Structural strength. Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.6 shall be cause for rejection of the thruster.

4.5.3.4 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.7 shall be cause for rejection of the lot.

4.5.3.5 Breakaway torque. Each body and holding ring shall be subjected to a breakaway torque. Failure of any unit to comply with the requirements of 3.5.2 shall be cause for rejection of the unit.

4.5.3.6 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.7 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.7. Failure of any unit in the inspection sample to comply with 3.10 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to application of protective finish the body and head shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

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4.6.3 Structural strength. The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.6.

4.6.4 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.7. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6.5 Breakaway torque. A torque wrench capable of measuring the torque requirements specified in 3.5.2 shall be used with an appropriate adapter to determine the adequacy of the joints. The thruster body shall be secured and the torque applied to the holding ring.

4.6.6 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.7 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8797633 and 8797634. Level B shall be in accordance with Drawings 8797633 and 9235819.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-921-5048.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

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b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special instruction requirements (see 4.3)

f. Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Preparing activity

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